

TED UNIVERSITY

CMPE 490 Introduction to Research in Computer Engineering

DriftWatch Research Statement

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DriftWatch: Predicting the Performance Degradation of Deployed Models from Unlabeled Input Data

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Problem: Once a deep learning model is deployed, the appearance of the data reaching it changes over time: different capture devices, resolutions, compression settings and acquisition conditions. The model produces output that appears completely normal and raises no errors with this input distribution shift, yet its accuracy subtly declines. Detecting this silent degradation in the conventional way requires labeled data, but production data arrives unlabeled. In reality, teams are faced with two unappealing choices: either wait until a user reports an issue or pay for human annotation of a sample on a regular basis. The model may have been producing inaccurate results for months by that point.

Gap: Two gaps motivate this work. First, although it is well established that distribution shift harms model performance, the contribution of individual visual factors has not been isolated, because in real-world data these factors are confounded: a shift in source typically changes resolution, compression and noise simultaneously. Second, the effectiveness of current drift detection techniques is assessed based on whether they identify a distributional shift rather than whether the alteration they identify really impairs task performance. It has not been determined whether the two align, and there is cause for concern that they do not.

Proposed Approach: The research unfolds in two phases. A controlled degradation benchmark is created in the first step: individual degradation factors (noise, blur, JPEG compression, resolution loss, contrast shift, and geometric distortion) are applied separately at varying severity levels to a labeled test set. Since only the look of each sample is modified while its content stays intact, the current ground truth continues to be applicable and no further annotation is needed. This produces an accuracy–severity curve for each factor and a robustness profile for every model. In the second phase, a label-free drift detector obtains identical image quality and statistical characteristics from incoming data and compares them with a reference distribution using Kolmogorov–Smirnov, Wasserstein distance, and Maximum Mean Discrepancy, generating a drift score without labels and without executing the model. The benchmark from the initial phase provides what this score otherwise misses: a link between seen input variation and anticipated accuracy loss.

Research Question: Can label-free, input-side drift signals predict the performance degradation of a deployed deep learning model, and does weighting these signals by the per-factor impact measured in a controlled degradation benchmark improve that prediction? The hypothesis is that standard drift metrics capture perceptual change rather than task-relevant change, that the two correlate weakly, and that impact-weighting yields a stronger correlation with actual performance loss. Evaluation uses Spearman correlation between drift score and accuracy loss, AUROC for early-warning performance, and detection delay.

Expected Contribution: Three outputs are expected: a factor-resolved degradation benchmark reusable for comparing the robustness of new models; a performance-aligned evaluation of drift detection, which current work omits; and an impact-weighted drift score combining the two. A negative result for the weighting is itself reportable, since the alignment measurement stands independently.